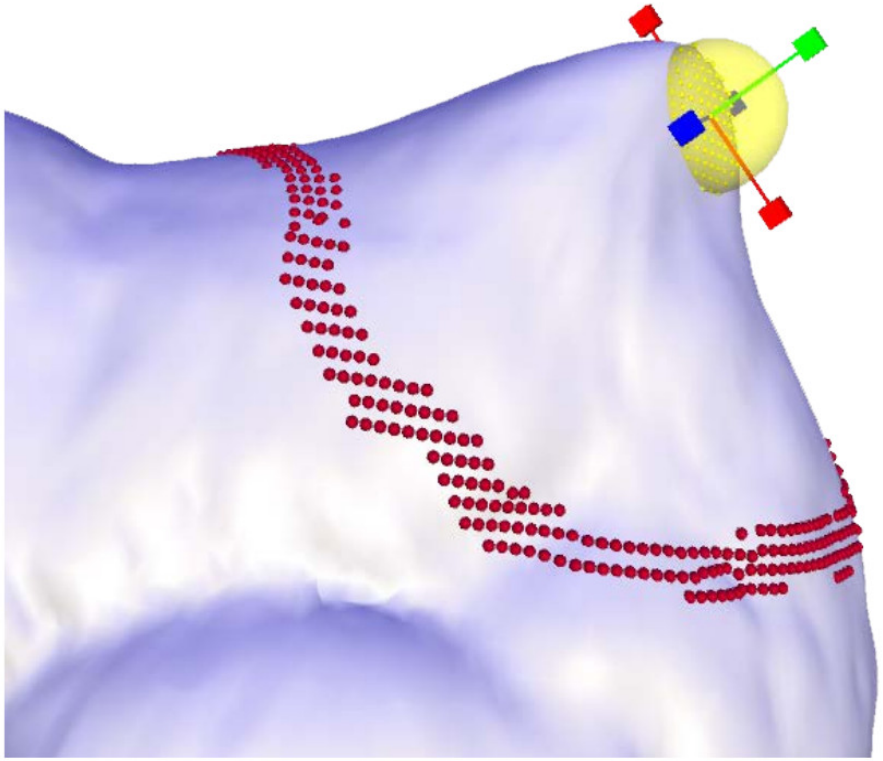
Advanced Topics in Computer Graphics II
Geometry Processing

Mesh Editing

December 4, 2024

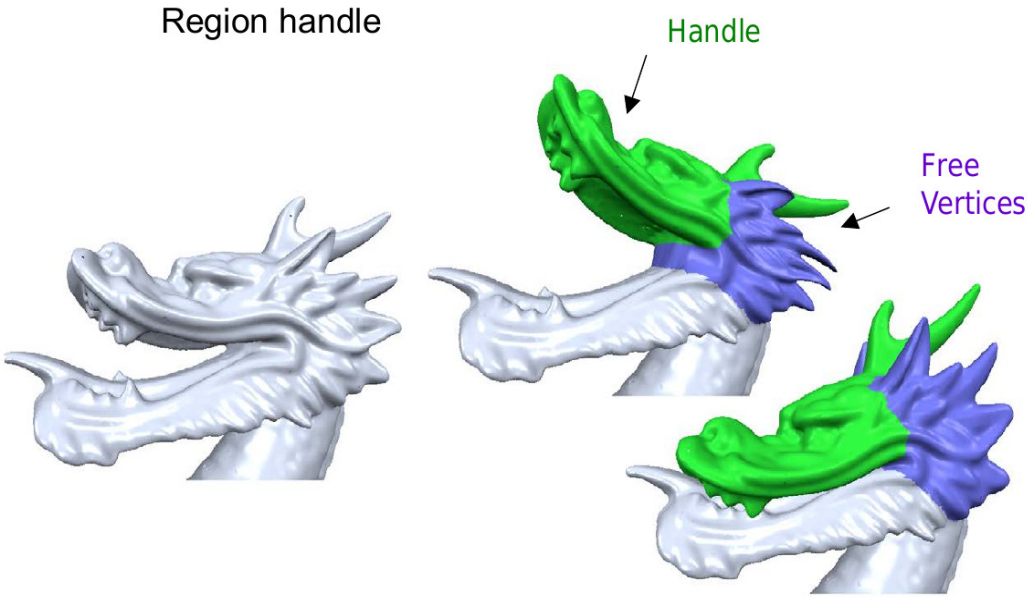
Motivation: What is Mesh Editing?

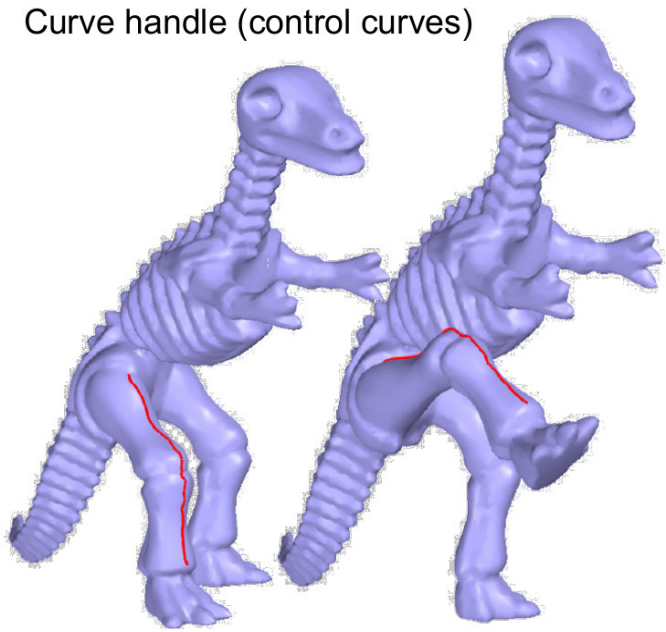
1. Start with a triangle mesh $\mathcal{M}$
2. Determine a submesh $\mathcal{R} \subset \mathcal{M}$ called the **Region of Interest (ROI)**
3. Determine **handles** within the ROI
4. Transform handles to the new pose
   $\rightarrow$ **Constraints**



$\rightarrow$ The algorithm moves all vertices in the ROI such that they satisfy the pose constrains and certain smoothness constraints are fulfilled

More examples:

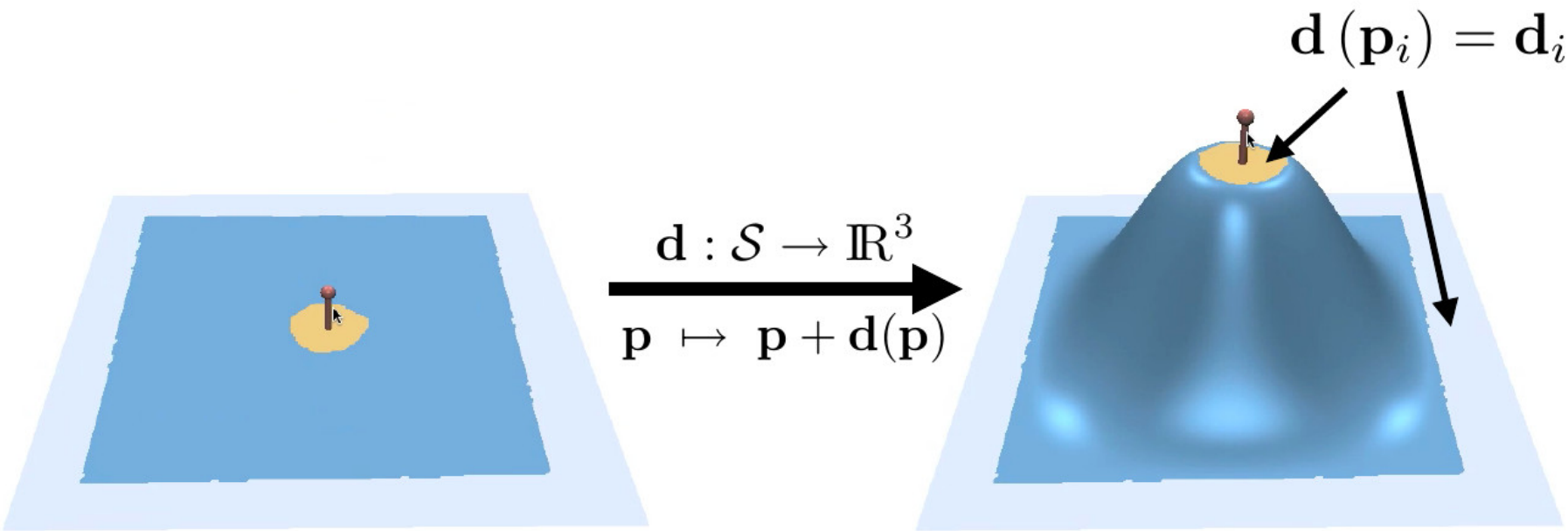




Idea: Model the mesh deformation by a displacement function[1]

$$\boldsymbol{d}\colon \Omega \subset \mathbb{R}^2 \to \mathbb{R}^3$$

and constrain the displacement at the handle and the fixed regions.



[1] O. Sorkine and M. Botsch. "Interactive Shape Modeling and Deformation". In: *Eurographics 2009 - Tutorials*. Ed. by K. Museth and D. Weiskopf. The Eurographics Association, 2009.

We want an intuitive deformation, so define a physically-based energy[2]

$$E = \int_{\Omega} k_s \left\| \boldsymbol{M}_1'(\boldsymbol{u}) - \boldsymbol{M}_1(\boldsymbol{u}) \right\|_F^2 + k_b \left\| \boldsymbol{M}_2'(\boldsymbol{u}) - \boldsymbol{M}_2(\boldsymbol{u}) \right\|_F^2 d\boldsymbol{u}$$

consisting of a stretching term, i.e. change of local distances, and a bending term, i.e. change of local curvature, each weighted with stiffness terms k_s, k_b.

Problem: The energy is non-linear and therefore hard to solve.

Idea: Linearize the energy and express it in terms of the displacement function

$$E = \int_{\Omega} k_s \left(\|\boldsymbol{d}_x(\boldsymbol{u})\|^2 + \|\boldsymbol{d}_y(\boldsymbol{u})\|^2 \right) + k_b \left(\|\boldsymbol{d}_{xx}(\boldsymbol{u})\|^2 + 2\,\|\boldsymbol{d}_{xy}(\boldsymbol{u})\|^2 + \|\boldsymbol{d}_{yy}(\boldsymbol{u})\|^2 \right) d\boldsymbol{u}$$

where $\boldsymbol{u} = (u_x, u_y)^T$ is a 2D coordinate and $\boldsymbol{d}_x = \frac{\partial}{\partial u_x}\boldsymbol{d}$ and $\boldsymbol{d}_y = \frac{\partial}{\partial u_y}\boldsymbol{d}$ the first partial derivatives. The second partial derivatives are defined in the same way.

[2] D. Terzopoulos et al. "Elastically deformable models". In: *ACM Siggraph Computer Graphics*. Vol. 21. 4. ACM. 1987, pp. 205–214.

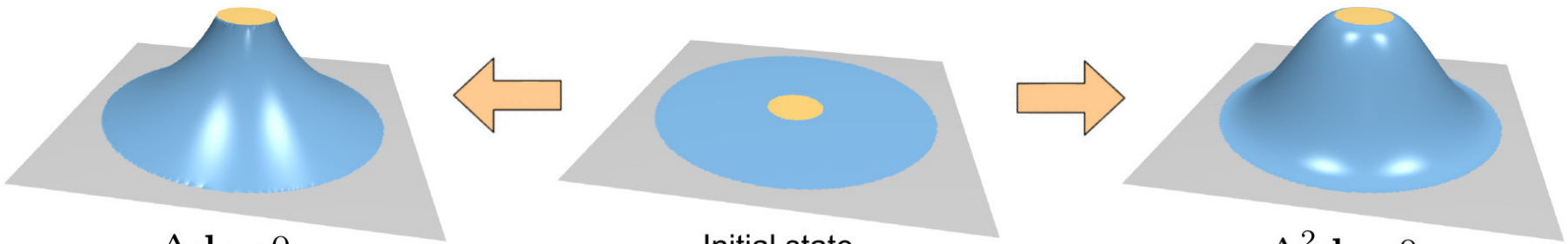
$$E = \int_{\Omega} k_s \left(\|\boldsymbol{d}_x(\boldsymbol{u})\|^2 + \|\boldsymbol{d}_y(\boldsymbol{u})\|^2 \right) + k_b \left(\|\boldsymbol{d}_{xx}(\boldsymbol{u})\|^2 + 2\,\|\boldsymbol{d}_{xy}(\boldsymbol{u})\|^2 + \|\boldsymbol{d}_{yy}(\boldsymbol{u})\|^2 \right) d\boldsymbol{u}$$

If we now apply variational calculus, we get the following Euler-Lagrange equation

$$-k_s\,\Delta \boldsymbol{d} + k_b\,\Delta^2 \boldsymbol{d} = \mathbf{0}$$

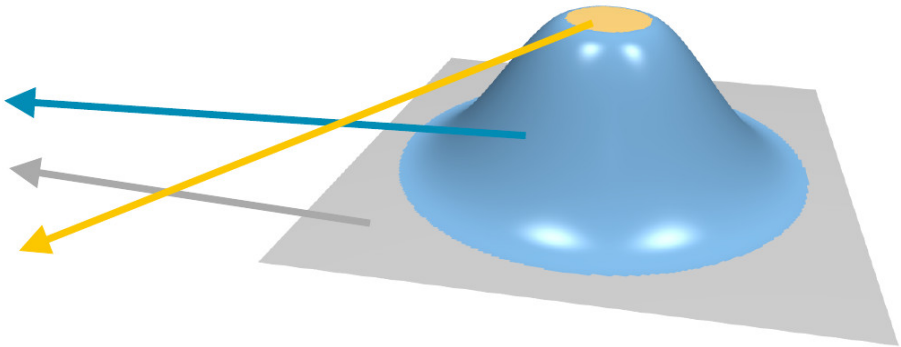
where

$$\Delta \boldsymbol{d} = \operatorname{div} \nabla \boldsymbol{v} = \boldsymbol{d}_{xx} + \boldsymbol{d}_{yy}$$
$$\Delta^2 \boldsymbol{d} = \Delta(\Delta \boldsymbol{d}) = \boldsymbol{d}_{xxxx} + 2\,\boldsymbol{d}_{xxyy} + \boldsymbol{d}_{yyyy}$$

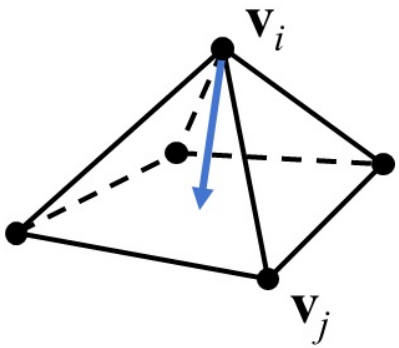


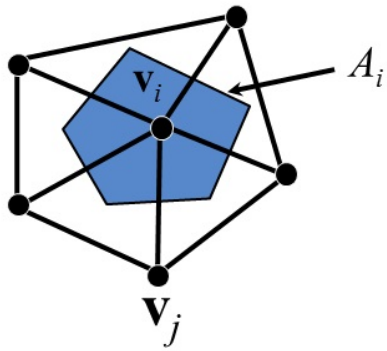
We obtain the final solution by solving a sparse linear equation system

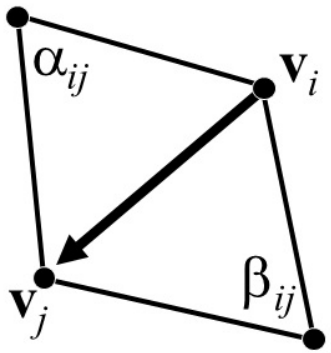
$$\begin{pmatrix} -k_s\,\Delta + k_b\,\Delta^2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{I} \end{pmatrix} \begin{pmatrix} \vdots \\ \boldsymbol{d}_i \\ \vdots \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{d}_j \end{pmatrix}$$



where the constraints for the fixed parts are needed to get a correct result.







The Laplace-Beltrami operator can be approximated using cotangent weights.

Given: A function $f : \Omega \to \mathbb{R}$ defined on the domain $\Omega \subset \mathbb{R}^2$.

Idea: Manipulate/deform the gradients[3][4] of f using a mapping $\boldsymbol{T}$:

$$\boldsymbol{g}(\boldsymbol{u}) \mapsto \boldsymbol{T}(\boldsymbol{g}(\boldsymbol{u})) \quad , \quad \boldsymbol{g}(\boldsymbol{u}) = \nabla f(\boldsymbol{u})$$

Find a new function $f^{'}$ whose gradient field best matches the target:

$$E[f^{'}] = \int_{\Omega} \left\| \nabla f^{'}(\boldsymbol{u}) - \boldsymbol{T}(\boldsymbol{g}(\boldsymbol{u})) \right\|_2^2 d\boldsymbol{u}$$

Applying variational calculus leads to the following Euler-Lagrange equation:

$$\Delta f^{'}(\boldsymbol{u}) = \operatorname{div} \boldsymbol{T}(\boldsymbol{g}(\boldsymbol{u}))$$

[3] Y. Yu et al. "Mesh editing with poisson-based gradient field manipulation". In: *ACM Transactions on Graphics (TOG)*. vol. 23. 3. ACM. 2004, pp. 644–651.

[4] M. Botsch et al. "Deformation Transfer for Detail-Preserving Surface Editing". In: *Vision, Modeling & Visualization*. 2006.

In order to apply this method to a mesh with vertices $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$, we need to know how to define the vertices in terms of the scalar filed f.

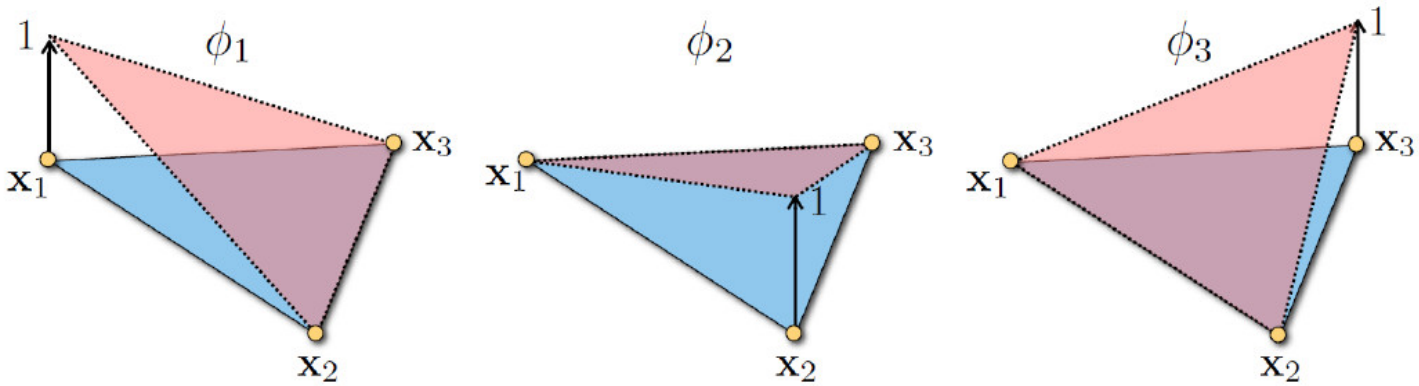
Consider the piece-wise linear coordinate function

$$\boldsymbol{v}(\boldsymbol{u}) = \sum_{i=1}^{n} \phi_i(\boldsymbol{u})\, \boldsymbol{v}_i$$

defining a vector field over the mesh. This vector field can be seen as a smooth generalization of the mesh meaning that

$$\forall \boldsymbol{u}_j \in \Omega \colon \boldsymbol{v}(\boldsymbol{u}_j) = \boldsymbol{v}_j \quad \Rightarrow \quad \phi_i(\boldsymbol{u}_j) = \delta_{ij}$$

This implies that the basis functions ϕ are hat functions and fulfill the above condition.



The gradient of this vector field is given by

$$\nabla \boldsymbol{v}(\boldsymbol{u}) = \sum_{i=1}^{n} \nabla \phi_i(\boldsymbol{u})\, \boldsymbol{v}_i^T$$

where the gradient of each basis function is constant per triangle. For a triangle with indices i, j, k, we can obtain them by encoding their properties into an equation system:

$$\begin{pmatrix} (\boldsymbol{v}_i - \boldsymbol{v}_k)^T \\ (\boldsymbol{v}_j - \boldsymbol{v}_k)^T \\ \boldsymbol{n}^T \end{pmatrix} \begin{pmatrix} \nabla \phi_i & \nabla \phi_j & \nabla \phi_k \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

- Inverse magnitude equal to the length h

$$\|\nabla \phi_i\| = \tfrac{1}{h} \Leftrightarrow \langle\, \nabla \phi_i \,|\, \boldsymbol{v}_i - \boldsymbol{v}_k \,\rangle = 1$$

- Perpendicular to the opposite edge $\boldsymbol{v}_j - \boldsymbol{v}_k$

$$\langle\, \nabla \phi_i \,|\, \boldsymbol{v}_j - \boldsymbol{v}_k \,\rangle = 0$$

- Perpendicular to the triangle normal $\boldsymbol{n}$

$$\langle\, \nabla \phi_i \,|\, \boldsymbol{n} \,\rangle = 0$$

Therefore the vector field gradient is also contant per triangle F_j.

$$\boldsymbol{g}_j := \nabla \boldsymbol{v}|_{F_j}$$

So we can build an operator $\boldsymbol{G} \in \mathbb{R}^{3|F| \times |V|}$ that computes the gradient of the coordinate function

$$\begin{pmatrix} \boldsymbol{g}_1 \\ \vdots \\ \boldsymbol{g}_{|F|} \end{pmatrix} = \boldsymbol{G} \begin{pmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_{|V|} \end{pmatrix}$$

and then manipulate the gradients to reflect our deformation:

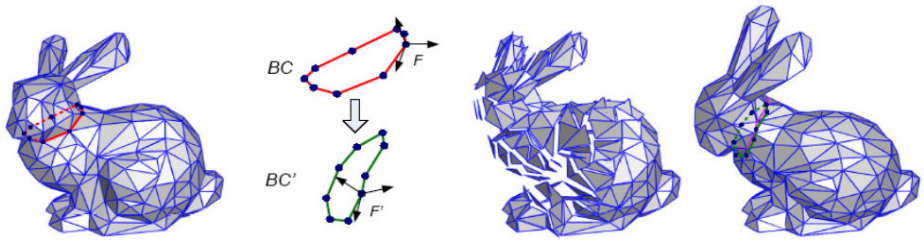
$$\boldsymbol{g}_j \mapsto \boldsymbol{T}_j(\boldsymbol{g}_j)$$

Finally, the new coordinates are obtained by solving the Euler-Lagrange equation

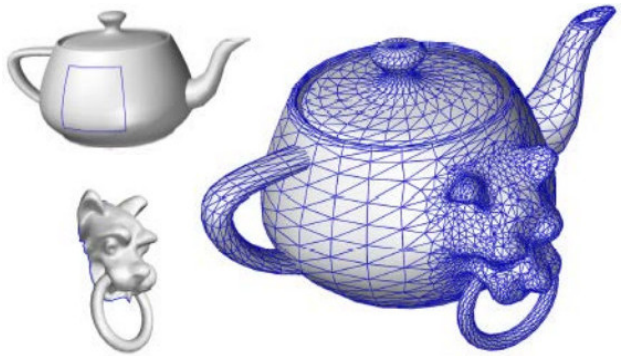
$$\underbrace{\boldsymbol{G}^T \boldsymbol{D} \boldsymbol{G}}_{\Delta} \begin{pmatrix} \boldsymbol{v}_1' \\ \vdots \\ \boldsymbol{v}_{|V|}' \end{pmatrix} = \underbrace{\boldsymbol{G}^T \boldsymbol{D}}_{\mathrm{div}} \begin{pmatrix} \boldsymbol{T}_j(\boldsymbol{g}_1) \\ \vdots \\ \boldsymbol{T}_j(\boldsymbol{g}_{|F|}) \end{pmatrix}$$

where $\boldsymbol{D}$ is a diagonal matrix containing the triangle areas.

Mesh Editing: Modify the gradients of the source mesh. Use the old mesh topology and the new gradient values to get a new set of vertex positions.

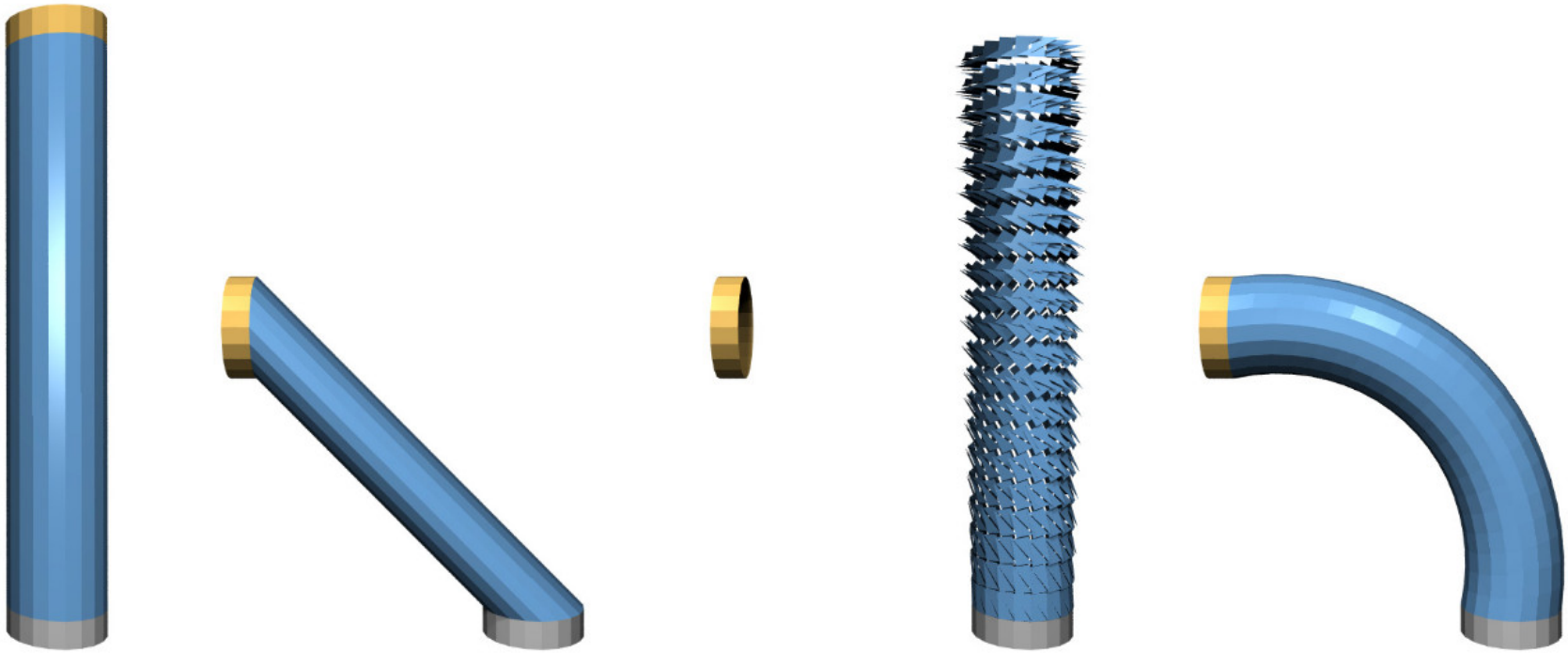


Mesh Merging: Link the two topologies together. Propage the changes in gradient information at the boundaries.



Mesh Smoothing: Smooth the mesh normals. Alter the gradients to reflect the changed normals, then solve for new vertex positions.

Problem: The resulting deformation is not intuitive.



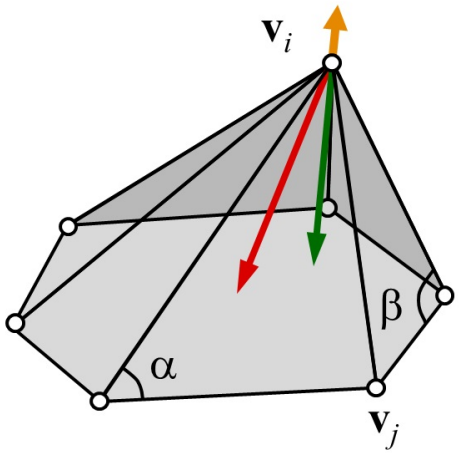
The approach tries to preserve the original mesh gradients with their orientation in the global coordinate system.

This ignores the fact that in the deformed surface the gradients should rotate as a consequence of the transformation of the vertices and thus the triangles.

Idea: Instead of manipulating per-face gradients, try to deform per-vertex Laplacians.

$$\delta(\boldsymbol{u}) \mapsto \boldsymbol{T}(\delta(\boldsymbol{u})) \quad , \quad \delta(\boldsymbol{u}) = \Delta f(\boldsymbol{u})$$

In case of meshes, the delta coordinates are direction vectors pointing into the negative normal direction.



Uniform Laplacian $L_u(v_i)$
Cotangent Laplacian $L_c(v_i)$
Mean curvature normal

For nearly equal edge lengths
Uniform ≈ Cotangent

Find a new function f' whose Laplacian field best matches the target:

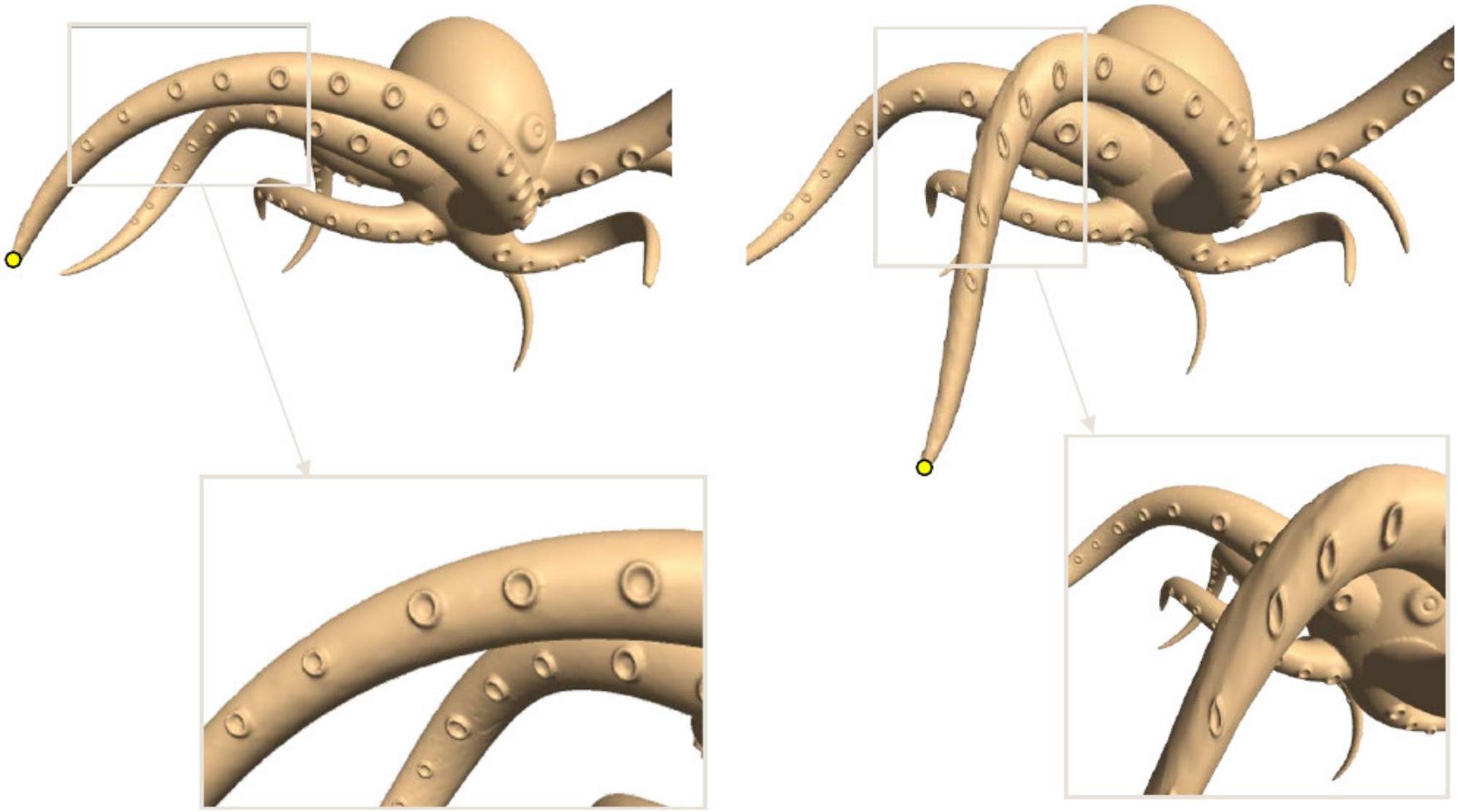
$$E[f'] = \int_{\Omega} \left\| \Delta f'(\boldsymbol{u}) - \boldsymbol{T}(\delta(\boldsymbol{u})) \right\|_2^2 d\boldsymbol{u}$$

Applying variational calculus leads to the following Euler-Lagrange equation:

$$\Delta^2 f'(\boldsymbol{u}) = \Delta \boldsymbol{T}(\delta(\boldsymbol{u}))$$

Compared to the gradient-based method, Laplacian-based editing leads to C^1-continuous instead of only C^0-continuous functions f'.

Problem: The reconstruction still attempts to preserve the original global orientation of the details



Definition

Given an $n \times n$ matrix $A = (a_{ij})$ over real or complex numbers, we define the exponential e^A of A as:

$$e^A = I_n + \sum_{p \geq 1} \frac{A^p}{p!} = \sum_{p \geq 0} \frac{A^p}{p!},$$

where $A^0 = I_n$.

The main question is: Why is this definition well-defined?

Theorem

Let $A = (a_{ij})$ be a $n \times n$ matrix. Define $\mu = \max\{|a_{ij}| \mid 1 \leq i, j \leq n\}$. If $A^p = (a_{ij}^{(p)})$, then:

$$|a_{ij}^{(p)}| \leq (n\mu)^p \text{ for all } i, j, 1 \leq i, j \leq n.$$

As a consequence of the proposition we get

- The n^2 series $\sum_{p \geq 0} \frac{a_{ij}^{(p)}}{p!}$ converge absolutely.
- The matrix $e^A = \sum_{p \geq 0} \frac{A^p}{p!}$ is a well-defined matrix.

Example: the exponential of a real skew-symmetric matrix:

$$A = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$$

First, observe that:

$$A = \theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad A^2 = -\theta^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Letting $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, we derive the following:

$$\begin{aligned} A^{4n} &= \theta^{4n} I_2, \\ A^{4n+1} &= \theta^{4n+1} J, \\ A^{4n+2} &= -\theta^{4n+2} I_2, \\ A^{4n+3} &= -\theta^{4n+3} J. \end{aligned}$$

Using the above inductive formula, the exponential of A is given by:

$$e^A = I_2 + \frac{\theta}{1!} J - \frac{\theta^2}{2!} I_2 - \frac{\theta^3}{3!} J + \frac{\theta^4}{4!} I_2 + \frac{\theta^5}{5!} J - \frac{\theta^6}{6!} I_2 - \frac{\theta^7}{7!} J + \cdots$$

Rearranging the terms in the series for e^A:

$$e^A = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots\right) I_2 + \left(\frac{\theta}{1!} - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right) J.$$

The series can be recognized as the power series for cosine and sine functions:

$$\begin{aligned} e^A &= \cos(\theta) I_2 + \sin(\theta) J \\ &= \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}. \end{aligned}$$

This shows that in this example e^A is a rotation matrix.

Note: This is a general fact. If A is a skew symmetric matrix, then e^A is an orthogonal matrix of determinant $+1$, i.e., a rotation matrix. Furthermore, every rotation matrix is of this form; i.e., the exponential map from the set of skew symmetric matrices to the set of rotation matrices is surjective.

General Linear Group

The set of real invertible $n \times n$ matrices forms a group under multiplication, denoted by $GL(n, \mathbb{R})$.

Special Linear Group

The subset of $GL(n, \mathbb{R})$ consisting of those matrices having determinant $+1$ is a subgroup of $GL(n, \mathbb{R})$, denoted by $SL(n, \mathbb{R})$.

Orthogonal and Special Orthogonal Groups

The set of real $n \times n$ orthogonal matrices forms a group under multiplication, denoted by $O(n)$. The subset of $O(n)$ consisting of those matrices having determinant $+1$ is a subgroup of $O(n)$, denoted by $SO(n)$. Matrices in $SO(n)$ are also called rotation matrices.

Vector Spaces

The set of real $n \times n$ matrices with null trace forms a vector space under addition, as does the set of skew-symmetric matrices.

Definition

The group $GL(n, \mathbb{R})$ is called the general linear group, and its subgroup $SL(n, \mathbb{R})$ is called the special linear group. The group $O(n)$ of orthogonal matrices is called the orthogonal group, and its subgroup $SO(n)$ is called the special orthogonal group (or group of rotations). The vector space of real $n \times n$ matrices with null trace is denoted by $\mathfrak{sl}(n, \mathbb{R})$, and the vector space of real $n \times n$ skew-symmetric matrices is denoted by $\mathfrak{so}(n)$.

Theorem

The exponential map $\exp : \mathfrak{so}(n) \to SO(n)$ *is well-defined and surjective.*[5]

[5] J. Q. Gallier and J. Quaintance. *Differential geometry and lie groups*. Vol. 12. Springer, 2020.

Consider a 3x3 real skew-symmetric matrix A[6]:

$$A = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}$$

Let $\theta = \sqrt{a^2 + b^2 + c^2}$ and define matrix B as:

$$B = \begin{pmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{pmatrix}$$

Theorem

(Rodrigues's Formula, 1840): The exponential map $\exp : \mathfrak{so}(3) \to SO(3)$ *is given by:*

$$e^A = \cos(\theta) I_3 + \frac{\sin(\theta)}{\theta} A + \frac{1 - \cos(\theta)}{\theta^2} B$$

Or, equivalently, by:

$$e^A = I_3 + \frac{\sin(\theta)}{\theta} A + \frac{1 - \cos(\theta)}{\theta^2} A^2$$

if $\theta \neq 0$, *with* $e^{0_3} = I_3$.

[6] J. O. Gallier and J. Quaintance. *Differential geometry and lie groups*. Vol. 12. Springer, 2020.

First observe that $A^2 = -\theta^2 I_3 + B$, since

$$A^2 = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} = \begin{pmatrix} -c^2 - b^2 & ba & ca \\ ab & -c^2 - a^2 & cb \\ ac & cb & -b^2 - a^2 \end{pmatrix}$$

$$= \begin{pmatrix} -a^2 - b^2 - c^2 & 0 & 0 \\ 0 & -a^2 - b^2 - c^2 & 0 \\ 0 & 0 & -a^2 - b^2 - c^2 \end{pmatrix} + \begin{pmatrix} a^2 & ba & ca \\ ab & b^2 & cb \\ ac & cb & c^2 \end{pmatrix}$$

$$= -\theta^2 I_3 + B,$$

and that $AB = BA = 0$.

From the above, deduce that $A^3 = -\theta^2 A$, and for any $k \geq 0$,

$$A^{4k+1} = \theta^{4k} A, \quad A^{4k+2} = \theta^{4k} A^2, \quad A^{4k+3} = -\theta^{4k+2} A, \quad A^{4k+4} = -\theta^{4k+2} A^2.$$

Then prove the desired result by writing the power series for e^A and regrouping terms so that the power series for $\cos(\theta)$ and $\sin(\theta)$ show up. In particular,

$$e^A = I_3 + \sum_{p \geq 1} \frac{A^p}{p!}$$

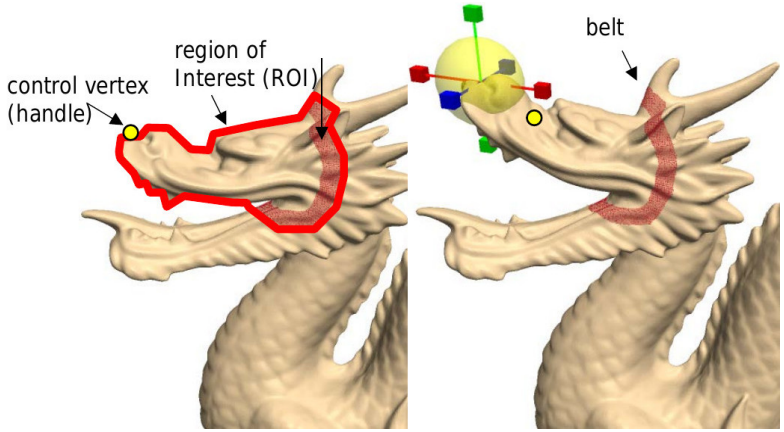
$$= I_3 + \sum_{p \geq 0} \frac{A^{2p+1}}{(2p+1)!} + \sum_{p \geq 1} \frac{A^{2p}}{(2p)!}$$

$$= I_3 + \sum_{p \geq 0} \frac{(-1)^p \theta^{2p}}{(2p+1)!} A - \sum_{p \geq 1} \frac{(-1)^{p-1} \theta^{2(p-1)}}{(2p)!} A^2$$

$$= I_3 + \frac{\sin(\theta)}{\theta} A + (1 - \cos(\theta)) \frac{A^2}{\theta^2}.$$

The above formulae are the well-known formulae expressing a rotation of axis specified by the vector (a, b, c) and angle θ. Since the exponential is surjective, it is possible to write down an explicit formula for its inverse (but it is a multivalued function!).

Observation: Gradient- and Laplacian-based Editing assumes that deformation of the whole gradient and Laplacian field is known. However, the user only wants to modify a specific region (ROI).



Idea: Solve for both the deformed mesh $\boldsymbol{V}'$ and the local transformations $\boldsymbol{T}_i$[7]:

$$E[\boldsymbol{V}'] = \sum_{i=1}^{n} \left\| \Delta \boldsymbol{v}_i' - \boldsymbol{T}_i(\delta_i) \right\|_2^2$$

Since the transformations $\boldsymbol{T}_i$ depend on the new mesh $\boldsymbol{V}'$, they are implicitly optimized.

[7] O. Sorkine, D. Cohen-Or, et al. "Laplacian surface editing". In: *Proceedings of the 2004 Eurographics/ACM SIGGRAPH symposium on Geometry processing*. ACM. 2004, pp. 175–184.

Each transformation $\boldsymbol{T}_i$ should map the vertex $\boldsymbol{v}_i$ and its neighbors to the deformed unknown vertices:

$$\left\| \boldsymbol{T}_i \, \boldsymbol{v}_i - \boldsymbol{v}_i^{'} \right\|_2^2 + \sum_{\boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{v}_i)} \left\| \boldsymbol{T}_i \, \boldsymbol{v}_j - \boldsymbol{v}_j^{'} \right\|_2^2$$

To avoid undesired solutions when minimizing this energy, we further constrain the transformations (in homogeneous coordinates) such that they satisfy the following conditions:

- Isotropic Scales $s_i \in \mathbb{R}$
- Rotations $\boldsymbol{\omega}_i \in \mathbb{R}^3$
- Translations $\boldsymbol{t}_i \in \mathbb{R}^3$

$$\boldsymbol{T}_i = \begin{pmatrix} s_i \, \exp\left[\boldsymbol{\omega}_i\right]_\times & \boldsymbol{t}_i \\ \boldsymbol{0} & 1 \end{pmatrix} \in \mathbb{R}^{4\times 4}$$

Rotation matrices $\boldsymbol{R}_i \in \mathbb{R}^{3\times 3}$ have 3 degrees of freedom (DoF) and can be efficiently represented as 3D vectors $\boldsymbol{\omega}_i$.

$$[\boldsymbol{\omega}_i]_\times = \begin{pmatrix} 0 & -\omega_{i,z} & \omega_{i,y} \\ \omega_{i,z} & 0 & -\omega_{i,x} \\ -\omega_{i,y} & \omega_{i,x} & 0 \end{pmatrix} \quad , \quad \forall \boldsymbol{x} \in \mathbb{R}^3 \colon [\boldsymbol{\omega}_i]_\times \boldsymbol{x} = \boldsymbol{\omega}_i \times \boldsymbol{x}$$

Here, $[\boldsymbol{\omega}_i]_\times$ denotes the cross-product matrix.

Assuming small rotation angles $\|\boldsymbol{\omega}_i\|_2$, we can linearize the rotation to obtain

$$s_i \, \exp\left[\boldsymbol{\omega}_i\right]_\times \approx s_i \left(\boldsymbol{I} + \left[\boldsymbol{\omega}_i\right]_\times\right) = s_i \, \boldsymbol{I} + \left[\boldsymbol{h}_i\right]_\times \quad , \quad \boldsymbol{h}_i = s_i \cdot \boldsymbol{\omega}_i$$

The scale factor s_i is moved into the rotation vector $\boldsymbol{\omega}_i$ to make the matrix linear in the unknown parameters:

$$\boldsymbol{T}_i = \begin{pmatrix} s_i & -h_{i,z} & h_{i,y} & t_{i,x} \\ h_{i,z} & s_i & -h_{i,x} & t_{i,y} \\ -h_{i,y} & h_{i,x} & s_i & t_{i,z} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Finally, we plug this into the energy function

$$\left\| \boldsymbol{T}_i \, \boldsymbol{v}_i - \boldsymbol{v}_i^{'} \right\|_2^2 + \sum_{\boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{v}_i)} \left\| \boldsymbol{T}_i \, \boldsymbol{v}_j - \boldsymbol{v}_j^{'} \right\|_2^2$$

and rewrite it to a linear least squares problem.

Therefore, we can write our problem as

$$\left\| \boldsymbol{A}_i \begin{pmatrix} s_i \\ \boldsymbol{h}_i \\ \boldsymbol{t}_i \end{pmatrix} - \boldsymbol{b}_i \right\|_2^2$$

where

$$\boldsymbol{A}_i = \begin{pmatrix} & \vdots & \\ \boldsymbol{v}_k & [\boldsymbol{v}_k]_\times & \boldsymbol{I} \\ & \vdots & \end{pmatrix} \in \mathbb{R}^{3|\mathcal{I}| \times 7} \quad , \quad \boldsymbol{b}_i = \begin{pmatrix} \vdots \\ \boldsymbol{v}_k' \\ \vdots \end{pmatrix} \in \mathbb{R}^{3|\mathcal{I}|}$$

and $k \in \mathcal{I} = \{i\} \cup \{j \mid \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{v}_i)\}$. Here, $\boldsymbol{A}_i$ only depends on the given vertices $\boldsymbol{v}_i$ whereas $\boldsymbol{b}_i$ contains the unknown vertices $\boldsymbol{v}_i'$. Therefore, the solution

$$\begin{pmatrix} s_i \\ \boldsymbol{h}_i \\ \boldsymbol{t}_i \end{pmatrix} = \underbrace{\left(\boldsymbol{A}_i^T \boldsymbol{A}_i\right)^{-1} \boldsymbol{A}_i}_{=:\boldsymbol{X}_i} \boldsymbol{b}_i$$

is obtained by applying the linear function $\boldsymbol{X}_i$ to the unknown vertices.

The transformations $\boldsymbol{T}_i$ are not explicitly computed. Instead, we plug its linear approximation and solution into the original energy function

$$E[\boldsymbol{V}^{'}] = \sum_{i=1}^{n} \left\| \Delta \boldsymbol{v}_i^{'} - \boldsymbol{T}_i(\delta_i) \right\|_2^2$$

to obtain

$$E[\boldsymbol{V}^{'}] = \sum_{i=1}^{n} \left\| \Delta \boldsymbol{v}_i^{'} - \boldsymbol{Y}_i(\delta_i)\, \boldsymbol{v}_i^{'} \right\|_2^2$$

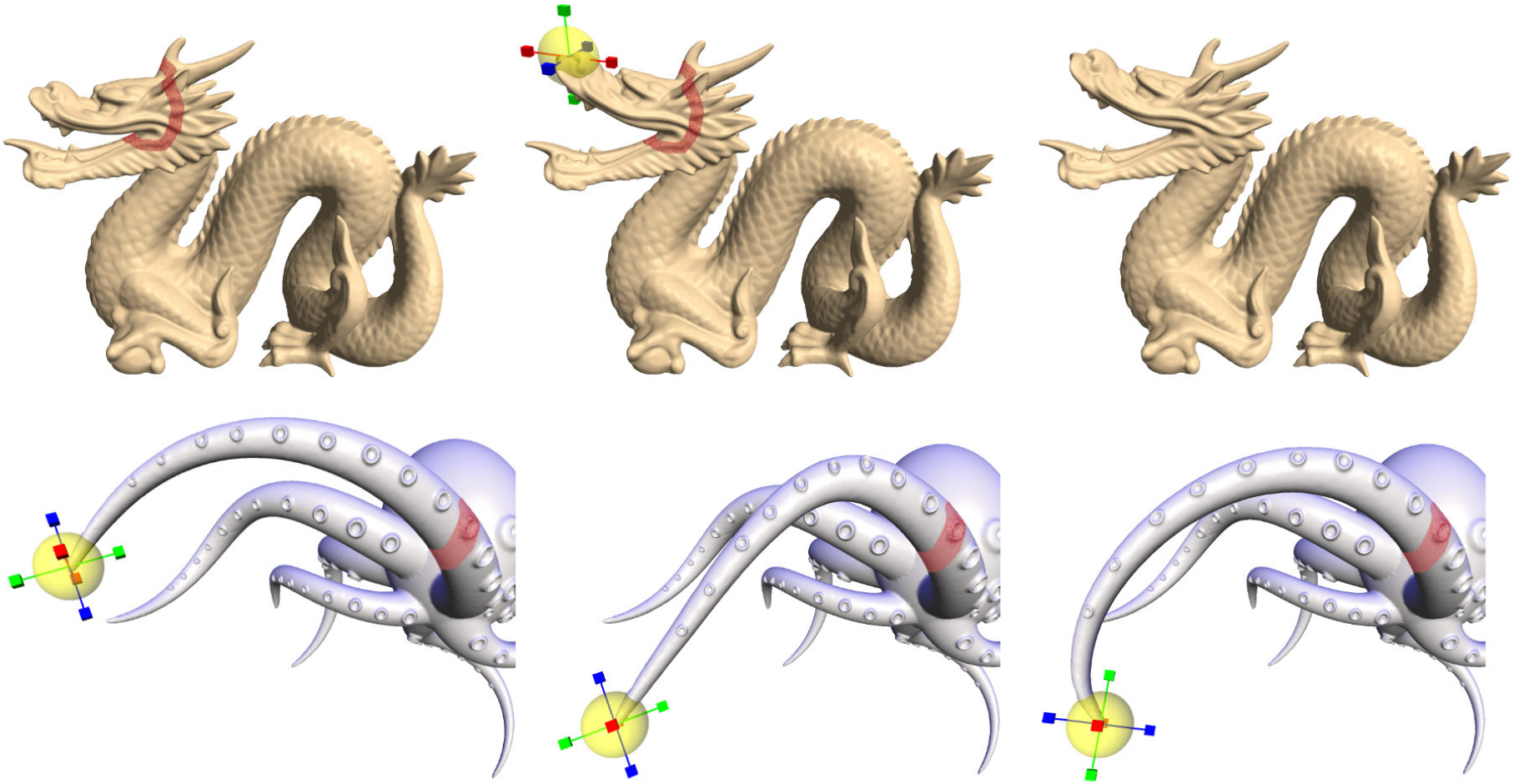
where $\boldsymbol{Y}_i(\delta_i)$ is the matrix that encodes the transformation and the delta coordinates. The fixed points $\boldsymbol{u}_l$ are added as a soft constraint into the energy function:

$$E[\boldsymbol{V}^{'}] = \sum_{i=1}^{n} \left\| \Delta \boldsymbol{v}_i^{'} - \boldsymbol{Y}_i(\delta_i)\, \boldsymbol{v}_i^{'} \right\|_2^2 + \sum_{l \in \mathcal{C}} \left\| \boldsymbol{v}_l^{'} - \boldsymbol{u}_l \right\|_2^2$$

Note: In case of too large rotation angles, an approximated reconstruction[8] can be computed and then refined using this technique.

[8] Y. Lipman et al. "Differential coordinates for interactive mesh editing". In: *Shape Modeling Applications, 2004. Proceedings*. IEEE. 2004, pp. 181–190.

Results:



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